EventCode: Event-Based Barcode Identification and Paired Evaluation against RGB Cameras

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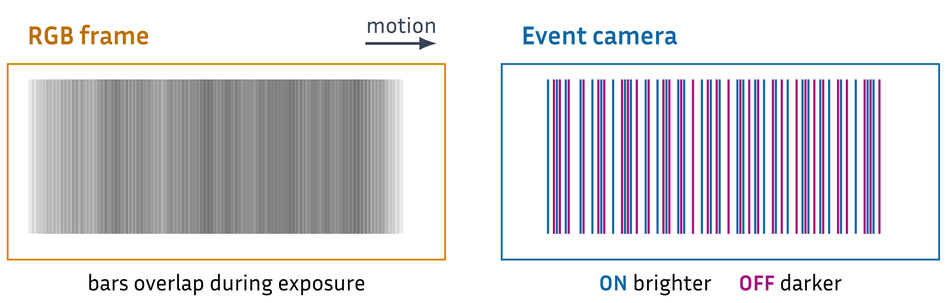
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Abstract

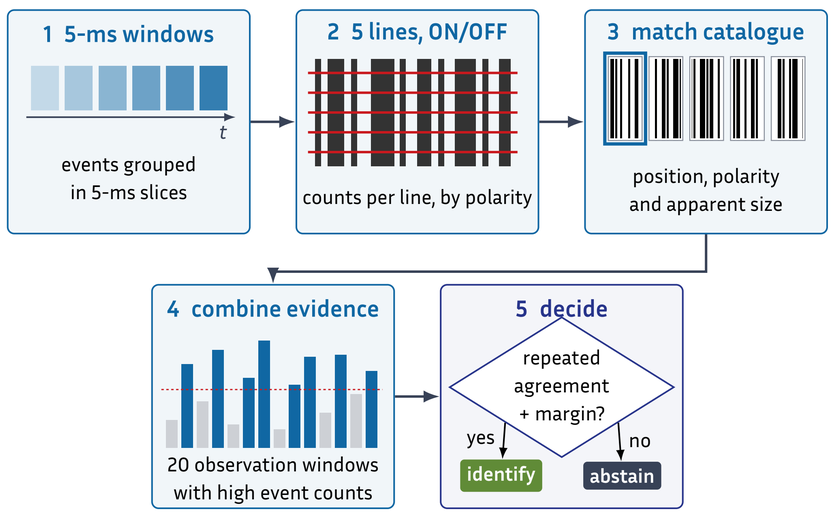
Fast motion blurs barcodes in RGB images, and a shorter exposure reduces blur but collects less light. Event cameras report per-pixel brightness changes with microsecond timing, but conventional barcode decoders cannot read this stream. We present EventCode, which identifies moving Code 128-B barcodes directly from events without reconstructing intensity video. EventCode counts ON and OFF events along five lines in 5-ms windows, matches the resulting profiles against a ten-label catalogue, and accumulates evidence before it identifies a label or abstains. We compared it with RGB decoding on 160 replayable paired passages, covering eight velocities and two RGB settings, on a shared 59.98-Hz LCD. Against 15-fps, long-exposure RGB, EventCode correctly identified 10/10 passages at 1200 px/s while RGB decoded 1/10 (exact McNemar $p = 0.0039$); 60-fps, short-exposure RGB correctly decoded all 80 moving passages and remained more reliable overall.

1 Problem: motion blur vs brightness changes



A barcode moving during the exposure blurs in RGB images[1]. Event-camera pixels instead report brightness changes with location, microsecond timestamp and polarity[2,3], a stream that conventional decoders cannot read.

2 EventCode: identifying labels from events

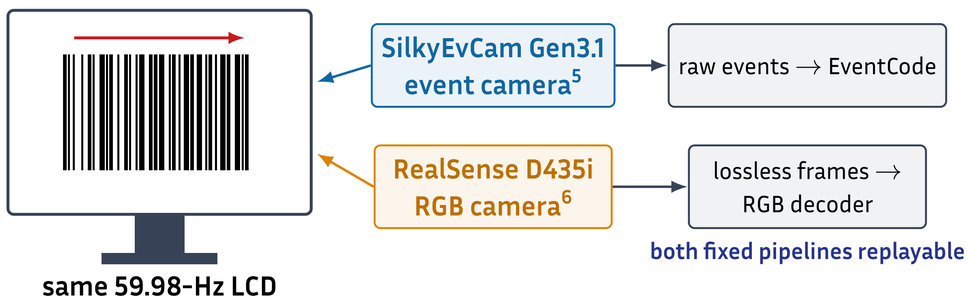


3 Closed catalogue with distractors



All ten are valid Code 128-B patterns with a check character[4]. EventCode identifies registered labels and does not decode arbitrary contents.

4 Paired, replayable setup



RGB baseline. The decoder samples eleven intensity scan lines per frame and accepts a payload only with a valid Code 128-B checksum.

RGB setting	Frame rate	Exposure control	Passages
Short exposure	60 fps	50	8 velocities × 10 reps
Long exposure	15 fps	500	8 velocities × 10 reps

Scoring separates correct identifications, wrong labels and abstentions, with Wilson confidence intervals[7] and paired exact McNemar tests[8].

STUDY HIGHLIGHTS

160 paired passages — 2 RGB settings, 80 each

8 display velocities — from 600 to 3000 px/s

10 catalogue labels — 4 shown, 6 distractors

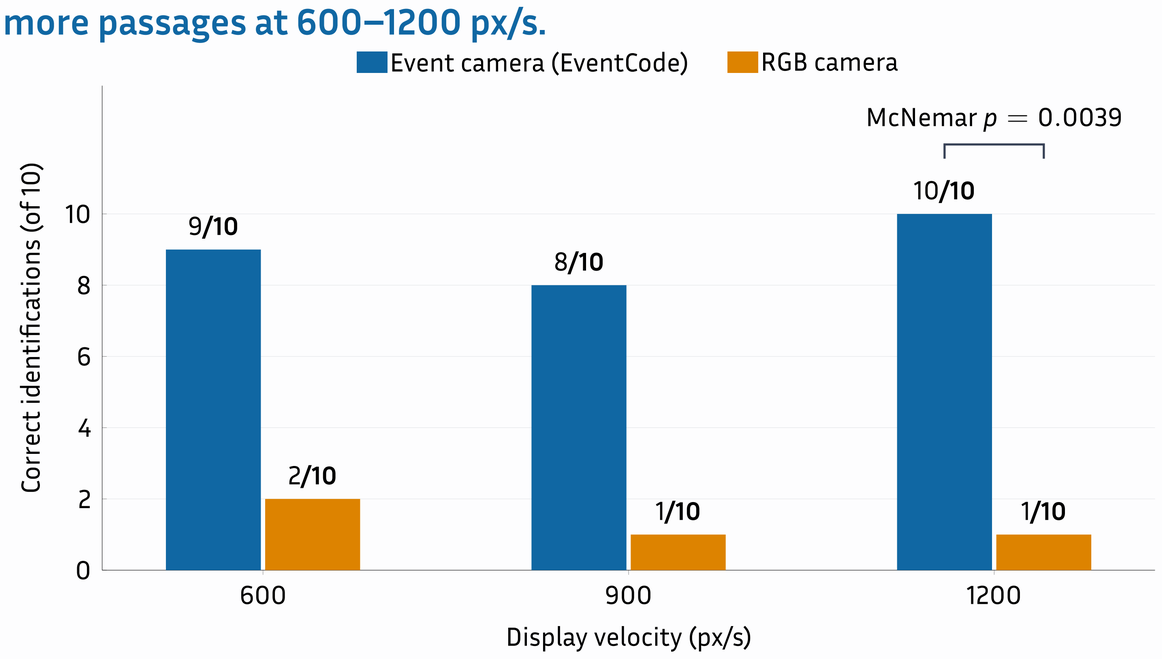
HEADLINE RESULT AT 1200 px/s

10/10 vs 1/10

correct identifications, EventCode vs 15-fps, long-exposure RGB (McNemar $p = 0.0039$)

5 Results

Against 15-fps, long-exposure RGB, EventCode correctly identifies more passages at 600–1200 px/s.



60-fps, short-exposure RGB correctly decoded all 80 moving passages.



one dot per passage: 8 velocities × 10 repetitions

Controls and caveats

Controls	static barcodes stayed RGB-readable in both settings, supporting a motion-related explanation; blank and black screens were also tested
Speed trend	no consistent decrease with speed; discrete LCD updates complicate it
Scope	compares the tested systems and settings, not isolated sensors

6 Take-home and next steps

EventCode identifies moving barcodes directly from events, with an advantage under the tested 15-fps, long-exposure RGB setting; 60-fps, short-exposure RGB remained more reliable overall. **Next:** physically moving printed barcodes, hardware-synchronised acquisition and unrestricted decoding.

References

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Code, contact and funding

Code

Contact

PRIMI Performance in Robots Interaction via Mental Imagery

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